Inferential statistics

(not for statisticians!)

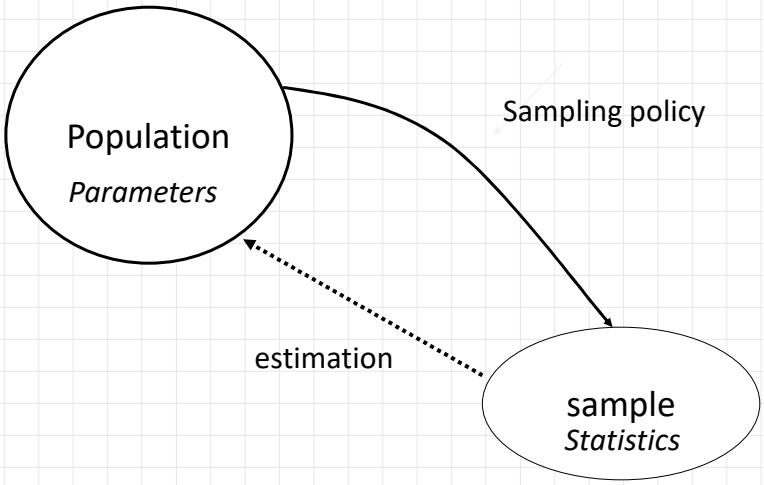
Inferential Statistics

Sampling

Problem: how to collect only a limited number of data, a sample, and through their analysis reach general conclusions, which can be extended to the whole population.

To reach these conclusions, inference must be used: *the ability to draw general conclusions* (about population or universe) *using only a limited number of variable data* (sample).

Statistical estimation



Gaussian distribution

If a sampling operation is repeated 20 times from the same population, each time with a different random sample, 20 different averages and 20 different ds will be obtained..

Key result: All these sample averages tend to assume a normal distribution, even if the population of origin is not normally distributed.

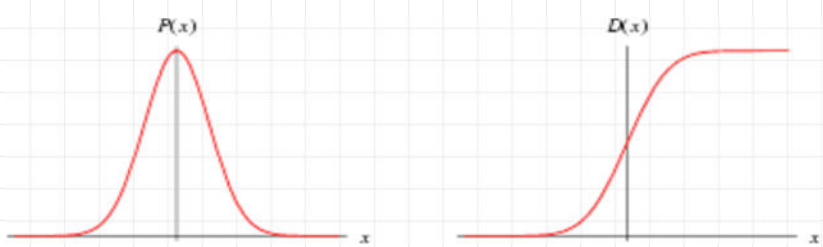
The random sampling process itself is a phenomenon that is normally distributed..

The normal distribution

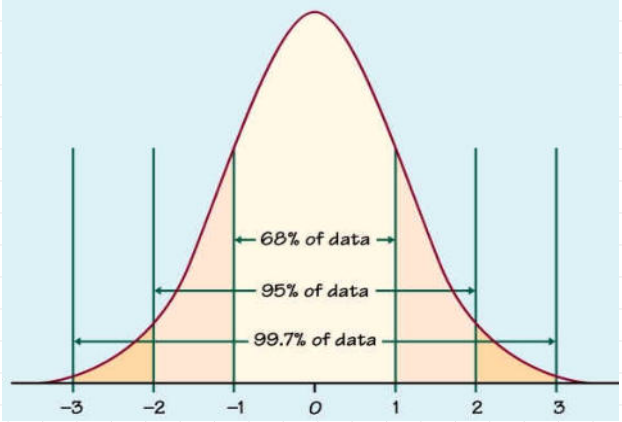
A normal distribution in a variable X with mean μ and variance σ^2 is a statistical distribution with probability function:

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

defined on the domain x ∈(∞, ∞).



Normal distribution (2)

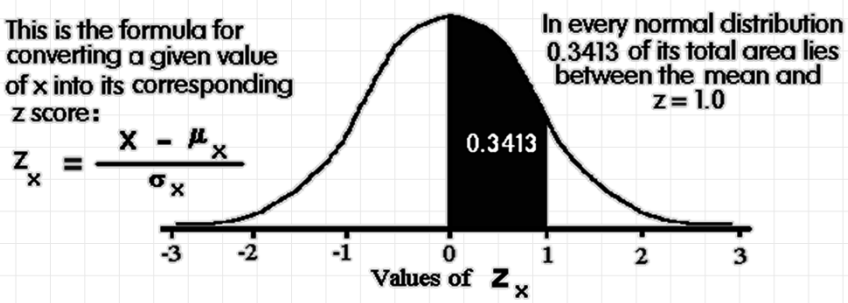


In a normal distribution :

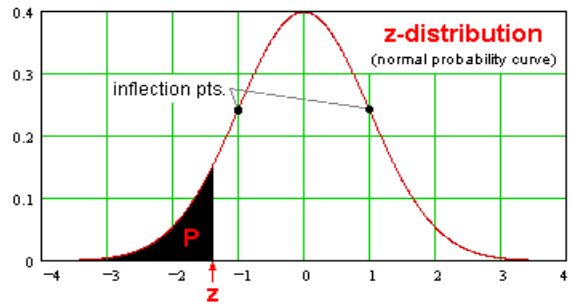
68.26% of cases are between **-1** e **+1** DS of the mean
95.46% of cases are between **-2** e **+2** DS of the mean
99.74% of cases are between **-3** e **+3** DS of the mean

z score

The ***z-score*** (*standard score*, *normal score*) It is a way of transforming every single value of a normal distribution into its ***standardized equivalent***, specifying how many ds the value is far from the population average.



z-distribution, probability



- The x-axis represents values. The ordinate represents the **probability density** of the values. The **area under the curve** represents the set of all possible cases, i.e. the total probability.
- Probabilities are never referred to a point, but to an interval, and represent the ratio of all cases within that range to the total number of cases.

Estimating intervals
Confidence interval (IC)

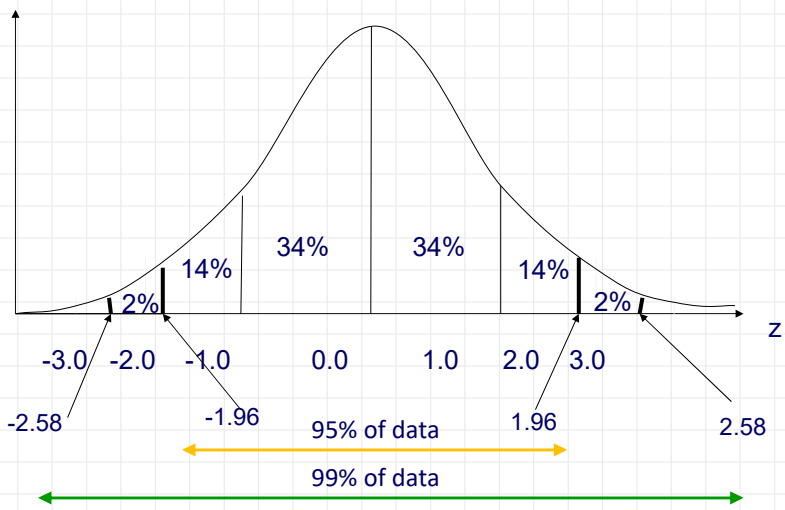
They provide a range of values within which we believe, with a certain level of confidence, that the true value falls

IC for population averages

$$95\%\ IC = \bar{x} \pm 1.96 \frac{Dev.St.}{\sqrt{n}}$$

$$99\%\ IC = \bar{x} \pm 2.58 \frac{Dev.St.}{\sqrt{n}}$$

Estimating intervals
Confidence interval (IC)



Statistical estimates and tests

We have experimental data and we want to describe them in aggregate with descriptive statistics and distributions.

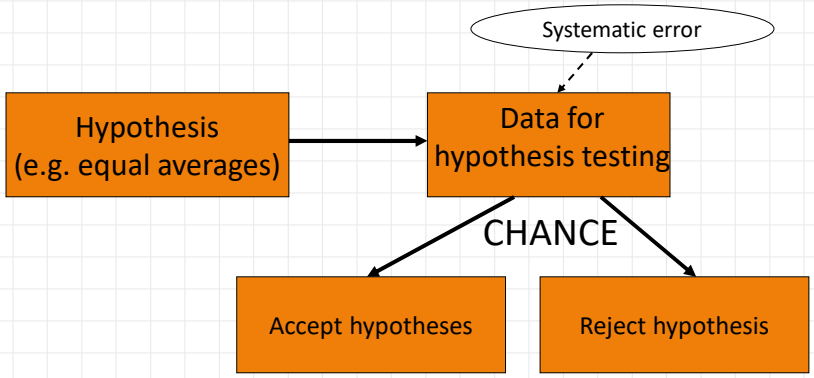
It is assumed that the sample was extracted from a given population.

We wonder how plausible this hypothesis is.

E.g. I have a machine that launches a *needsAttention* alarm every two days, do I have to worry?

E.g. I ran three different learning algorithms on my data, are the results equivalent or is one better than the others? Are they doing better than what I am doing by hand?

Statistical estimation: the role of chance



Random error (the chance) can be checked by managing statistical significance or confidence intervals.

Model selection

Given the evaluation of two machine learning methods on a dataset, which model do you choose? Clearly the model with the best performance, the one whose estimated performance (maximum accuracy or minimum error) when making predictions on unseen data is best.

But, is the difference in performance between two machine learning models real, or due to a statistical chance? Hypothesis testing gives an answer, considering

- Repeated Estimates. A sample of scores is required in order to calculate statistics, repeated training and testing of a given model on the same or different data.
- Distribution of Estimates. The sample of score estimates will have a distribution, possibly Gaussian. This will determine which tests can be used.
- Central Tendency. Performance will often be ultimately compared using a summary statistic (mean or median). The test will work on these.

Hypothesis testing

Statistical hypothesis testing is a logical-mathematical process that leads to the conclusion that the hypothesis of randomness cannot be rejected or that it can be rejected, through the calculation of probabilities of committing an error with these statements.

The hypothesis that the result obtained with the experimental data is ***due only to chance*** is called the **null hypothesis** and is denoted by **H0**.

It states that the differences between two or more groups are **essentially attributable to chance**.

Test

The question:

Assuming that the differences between groups of empirical observations are due to exclusively random factors, *what is the probability that among all the possible alternatives the situation described arises* from the data collected (or an even more extreme one)*?*

If this probability is (relatively) high, conventionally equal to or greater than 5%, the differences will be attributed to **purely random** factors (*acceptance* of the null hypothesis).

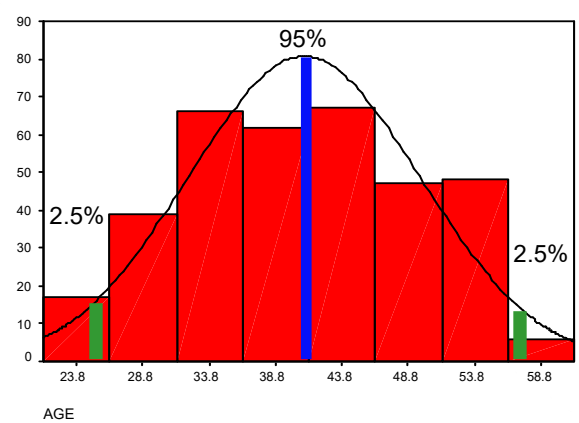
On the contrary, if the probability is low, say lower than 5%, it is accepted as likely that the differences are due to **non-random** factors (*non-acceptance* of the null hypothesis).

Example

If we flip a coin in the air 10 times consecutively, we have the following chances that it will come out headed. Question: is the coin rigged?

head	cross	tot. flips	p(%)	
10	0	10	0,10	
9	1	10	0,98	Null hypothesis rejected
8	2	10	4,39	
7	3	10	11,72	
6	4	10	20,51	
5	5	10	24,61	Null hypothesis accepted
4	6	10	20,51	
3	7	10	11,72	
2	8	10	4,39	
1	9	10	0,98	Null hypothesis rejected
0	10	10	0,10	

Hypothesis test: the p-value



The probability of having a value outside the green line range if the null hypothesis is true is < 5%

Hypothesis test: the p-value

p = probability of observing a value more extreme than that considered, if the null hypothesis is true

The lower the value of *p*, the greater the chance that the null hypothesis is an explanation of the data

In the example:

- Results outside the green lines: $p < 0.05$,
- Results inside the green lines: $p > 0.05$

Hypothesis testing: confidence intervals and significance

The value of the null hypothesis within the 95% interval ⟷ null hypothesis accepted: $p > 0.05$

The value of the null hypothesis outside the 95% interval ⟷ Null hypothesis rejected: $p < 0.05$

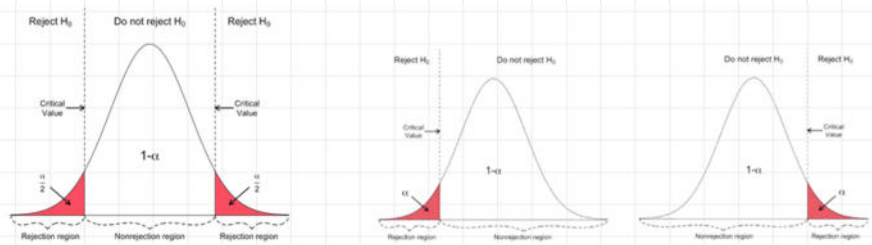
Parameters characterizing a test

Degrees of freedom

the number of points, elements or other units in the input data, which are free to vary,
Often equal to the number of observations minus 1.

Test on one or two tails

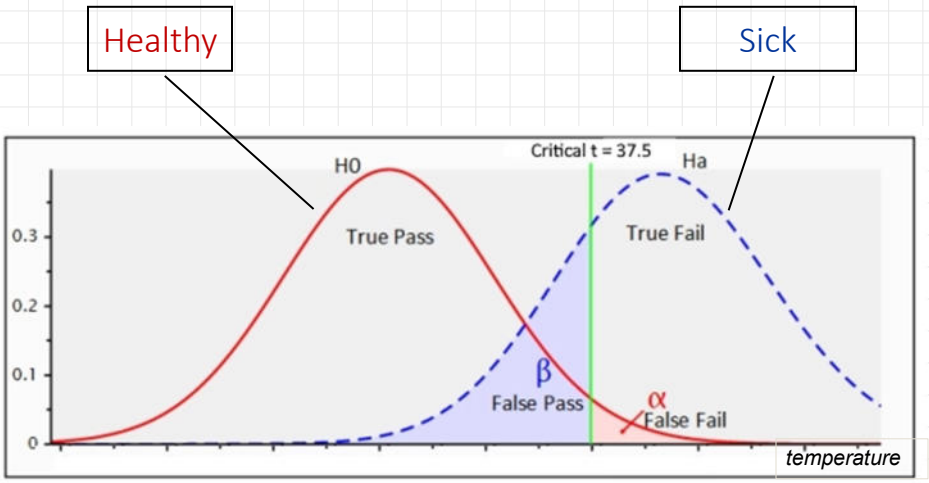
Tests on a tail are used for already oriented hypotheses
Two-tailed testing in all other cases



Type 1 and Type 2 errors

- You make a type 1 error when you reject a null hypothesis that is actually true.
- You make a type 2 error when you accept a null hypothesis that is in fact false.
- The probability of committing a type 1 error is denoted by α. The complementary situation (i.e. *not to be wrong*, accepting a null hypothesis that is true) has probabilities (1 - α).
- The probability of committing a type 2 error is denoted by β. The complementary situation (i.e. not to make mistakes, discarding a null hypothesis that is false) has probability (1-β).

Errors



Example: temperature to determine the state of health

Hypothesis testing: Type I and II errors

Decision	H_0 true / H_A false	H_0 true / H_A false
Accept H_0 / reject H_A	OK $p=1-\alpha$	Error: Type II (β) $p=\beta$
reject H_0 /accept H_A	Error: type I (α) $p=\alpha$	OK $p=1-\beta$

α: level of significance

$1-\beta$: power of the test

H0 and p value

Null Hypothesis, H0

- The null hypothesis is the statement that a difference does not exist between the distribution of the two samples of data; that any variance is seen due to noise or chance.
- In the case of machine learning, the null hypothesis is that the performance metrics are equal, that any small observed difference is not statistically significant.

p-value

- The probability of obtaining the observed data.
- If the p-value is below a given threshold (often 0.05), we can reject the null hypothesis and infer that the difference is statistically significant.

Errors

If the null hypothesis is false then some other hypothesis, H1, must be true. If we cannot specify this alternative hypothesis, it is not possible to determine the probability of committing a type II error.

It is often not possible to identify a single alternative hypothesis, so only the level of significance α is considered, without fixing β: it is considered more appropriate to guard against the most serious type of error, type I.

This assessment is motivated by the fact that while the rejection of H0 implies that the alternative hypothesis is true., Its acceptance implies that "there are not enough elements to reject it".

Procedure for an inference test

I – HYPOTHESIS
Null hypothesis, alternative hypothesis

II - DATA COLLECTION
Type of scale; Features of data distribution

III - CHOICE OF TEST
Based on the assumption, scale type and characteristics of the data

IV - TEST RESULT – PROBABILITY
Probability of obtaining that result, in the condition expressed by the null hypothesis

V – DECISION - CHOICE BETWEEN THE TWO HYPOTHESES
Probability α

VI – TEST AND DATA ANALYSIS FOR A NEW EXPERIMENT
Retrospective and a priori power, probability β

Testing: Test Types

Main distinction between tests *parametric* and *non-parametric*

Parametric Tests are based on assumptions on the distribution of population parameters. A normal (Gaussian) distribution is usually assumed. Parametric tests are more powerful (they're more likely to detect a statistically significant effect if one exists), but they can be *misleading* if the underlying assumptions are not satisfied.

Non-parametric tests They do not make assumptions about the distribution of the population (they are also called distribution free tests). **Usually based on the ranks of observations**, i.e. their order number rather than the observations themselves.

They have less power and are less flexible than parametric tests.

Testing: Test Types

Nonparametric tests are justified when:

1) the variables have obvious deviations from the normal (either they are strongly asymmetric or have more than one peak);
2) when the sample is too small to understand if there is a normal distribution of data;
3) when the observations are represented by ordinal rankings (es. severity of a disease from 1 to 4).

Tests

1) Relationships between variables:

Examples: correlation, regression, chi square.

2) Differences between variables:

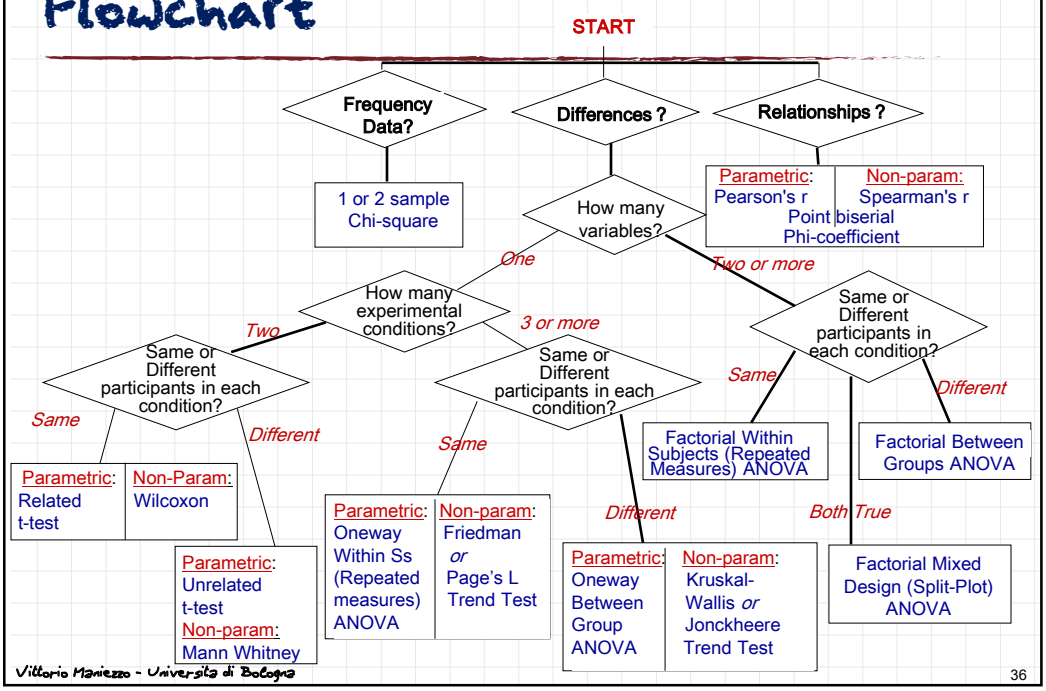
Es. tests on difference in effectiveness of different solution approaches

Examples: t-test; Analysis of Variance (ANOVA), Wilcoxon, ...

Paired or Unpaired Observations

If data was collected from the same subjects for each model, tests are "*paired*" (in ML, the data for all models are the same).
Data collected from two independent groups is referred to as "unpaired". This is common in scientific research, where one compares a control group with a treatment group.

Flowchart



Normality tests

It is used to check whether a sample of data is distributed normally or not.

Many ways to do it:

Graphical methods (simple, rude, unreliable):

- Symmetry
- Outlook of the distributionQ-Q plot

Analytical methods

More reliable tests should be used, such as the Kolmogorov-Smirnov, Anderson-Darling, or Shapiro-Wilk tests.

Normality test: method 1 (graphic)

Distribution symmetry analysis

Sufficiently large sample required, at least 50 points

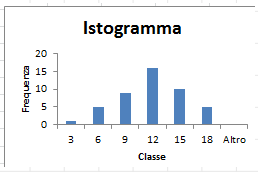
1. Calculate mean (A), median (M), range (R), and standard deviation (σ) of the sample.
2. A and M must be close together, < 1% of R. (normal distrib. is symmetrical, A = M). If very different, non-normal distribution.
3. Rule 68-95-99.7: in a normal distribution, 68% of the data are within σ from A, 95% within 2σ, 99.7% within 3σ.
4. If you pass 3 and 4 satisfied, the sample distribution may be normal.

Normality test: method 2 (graph)

Another immediate and rude way: comparison of the data histogram with the normal curve. It must have a single maximum, be symmetrical with degrading tails.

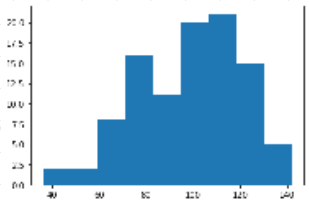
E.g. in Excel, you need a lot of data (> 100). You sort the data, group it arbitrarily (in "bins"). The lower limits of each bin are required. You need to activate the additional component "Data analysis", then choose histogram. Enter the data series in "input range" and the bins series in "Class range". Check "Output chart" and you have something like:

Classe	*Frequenza*
3	1
6	5
9	9
12	16
15	10
18	5
Altro	0

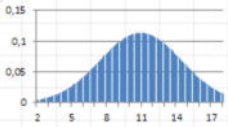


Es. in python

```
import numpy as np
import matplotlib.pyplot as plt
# generate data
np.random.seed(20)
data = 20 * np.random.randn(100) + 100
# histogram plot
nbins = 10
plt.hist(data,nbins-1)
plt.show()
```



Compare with a normal distribution with the same average and stddev:

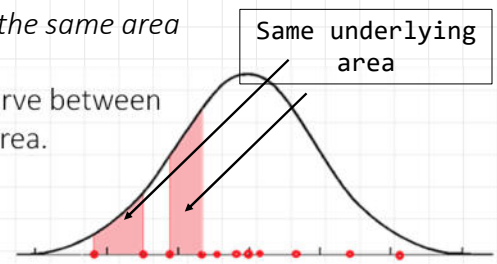


Normality test: Q-Q plot

Quantile-quantile plot. Compare the sample data with those that would have with the distribution of interest (here the Gaussian).

- Each sample data is coupled with a datum of the reference distribution (cumulative).
- The points are plotted in a scatter plot with the idealized value on the x-axis and the data sample on the y-axis. A perfect match for the distribution will be shown by a line of points at an angle of 45 degrees from the lower left of the diagram in the upper right.
- Point deviations from the line show a deviation from the expected distribution.

TEO: *Normally distributed data has the same area between two successive data*.
E.g., 7 points, the area under the curve between two next points is 1/7 of the total area.



Normality test: Q-Q plot

Example excel, data -4 -3 0.8 1.8 3.9 6.2 6.5 (sorted!)

Calculate n (7), average (1.74), and DS (3.85) of the DATA column, sorted.

- Add a column ***cum. z-score*** with *equispaced cumulative normal probabilities*. The average CDF is worth 0.5. The CDFs of the other data are centered on the mean (0.5) and then separated by 1/n (if there are even data, the two centers have an average of 0.5 and are separated by 1/n).
- Add a column ***cum.norm***, with z-data adapted to aample parameters:

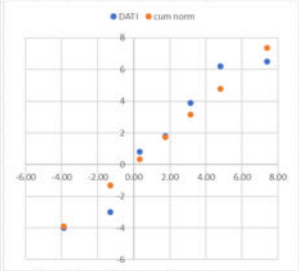
  INV.NORM.N(PROB, sample mean., sample dev.st.)
  (NORM.INV if English)

	DATA	cum z-score	cum norm
	-4	0.07	-3.89
	-3	0.21	-1.30
	0.8	0.36	0.33
	1.8	0.50	1.74
	3.9	0.64	3.15
	6.2	0.79	4.79
	6.5	0.93	7.38
average	1.74		
std	3.85		

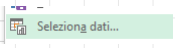
Normality test: Q-Q plot

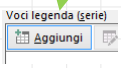
The test is performed by plotting in a graph (*scatter*) the values of the real data (DATA) against the normal ones (cum.norm)
In the graph, the data should approximate the line generated by the normal data.

	DATI	*cum z-score*	*cum norm*
	-4	0.07	-3.89
	-3	0.21	-1.30
	0.8	0.36	0.33
	1.8	0.50	1.74
	3.9	0.64	3.15
	6.2	0.79	4.79
	6.5	0.93	7.38
media	1.74		
std	3.85		

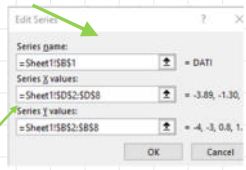


Graph: 1) insert graph 2) mouse dx select data 3) items add 4) edit series





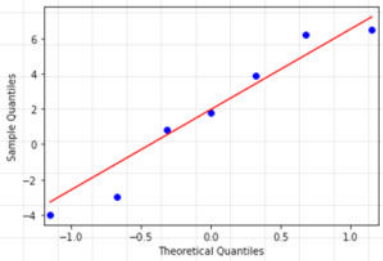
cum.norm



Q-Q plot, python

Predefined inside statsmodels

```
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.graphics.gofplots import qqplot
data = np.array([-4,-3,0.8,1.8,3.9,6.2,6.5])
# q-q plot
qqplot(data, line='q')
plt.show()
```



Normality test: statistical

Many possible tests, test different hypotheses:

- Shapiro-Wilk, among the most powerful
- D'Agostino K-squared, checks normality on the basis of skewness and kurtosis (skewness is a measure of symmetry, Kurtosis a measure of how much the tails are above or below normal ones).
- Anderson-Darling
- Chi-Square
- Kolmogorov-Smirnov, general test of distribution *Goodness of Fit*.
- . . .

Normality test

Example: Shapiro-Wilk

```
import numpy as np
from scipy.stats import shapiro
# generate data
np.random.seed(20)
data = 20 * np.random.randn(100) + 100
# normality test
alpha = 0.05
stat, p = shapiro(data)
print('Shapiro=%.3f, p=%.3f, alpha=%.3f' % (stat, p, alpha))
if p > alpha:
    print('Campione gaussiano (non rifiuto H0)')
else:
    print('Campione NON gaussiano (rifiuto H0)')
```

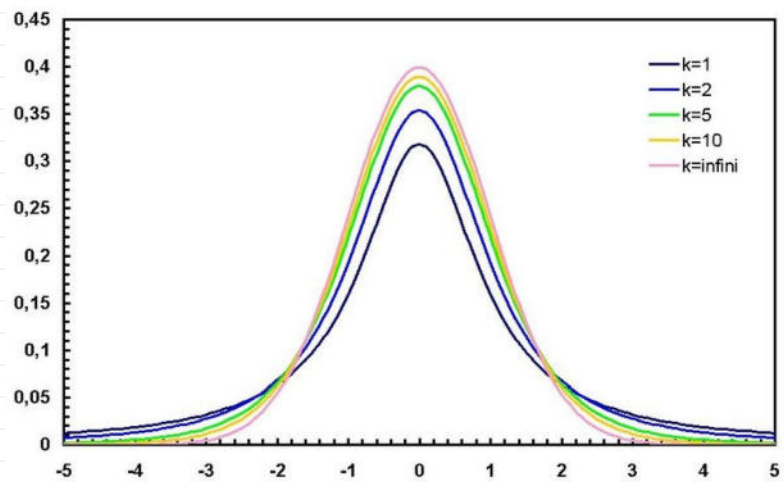
```
Shapiro=0.983, p=0.241, alpha=0.050
Gaussian sample (no rejection of H0)
```

Example: Kolmogorov-Smirnov (*compares with standard gaussian*)

```
from scipy.stats import kstest
stat, p = kstest(data,'norm')
print('Kolmogorov=%.3f, p=%.3f, alpha=%.3f' % (stat, p, alpha))
if p > alpha:
    print('Campione gaussiano (non rifiuto H0)')
else:
    print('Campione NON gaussiano (rifiuto H0)')
```

```
Kolmogorov=1.000, p=0.000, alpha=0.050
NON-Gaussian sample (reject H0)
```

Student's t distribution



Family of distributions varying k = degrees of freedom

t test

The *Student's t* test is a parametric test which is used to check if there has been an effect from an operation (coupled data, measuring before and after) or otherwise to check if two groups (samples) are different.

It compares two averages and it says if they are significantly different. A value of t is calculated, a function of the difference between means and variances. The higher t the more likely the averages are to be different.

Generally

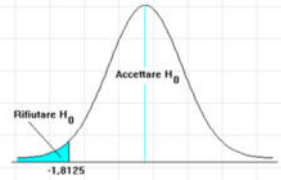
t = (Difference between means) / (group variability)

The specific formula depends on the type of test, there are many of them

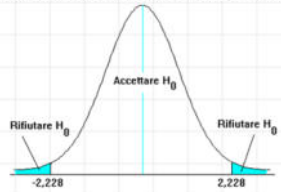
Greater variance
smaller t
More similar means

t-test one or two tails

In the one-tail test, the rejection zone is only on one side of the distribution (left when the sign is negative, right when it is positive)



In the two-tailed test, the waste zone is distributed on the two sides



The two-tailed test is more conservative (it is used when you have no idea about the possible results) while the one-queue test is more powerful

Nonparametric Tests: Ordinal Data

Independent groups

- Wilcoxon *rank-sum* (Mann-Whitney U): Comparison between ***two*** groups (ako not paramateric Student's t)
- Kruskal-Wallis H: Comparison among ***two or more*** groups

Related groups

- Wilcoxon *matched-pairs signed-rank test*: Comparison between ***two*** groups
- Friedman matched samples: Comparison between ***two or more*** groups

Wilcoxson

Wilcoxson tests assumptions:

- The observations from the two groups should be randomly selected from the target populations
- Observations are independent of each other
- Observations should be continuous or ordinal

Python (scipy):

```
import scipy.stats as stats
# perform two-sided test. You can use 'greater' or 'less' for one-sided test
res = stats.mannwhitneyu(x=a1, y=a2, alternative = 'two-sided')
print(res)
>>>MannwhitneyuResult(statistic=42.0, pvalue=0.08853377201422956)
```

Wilcoxon rank-sum in case of continuous distributions:

```
from scipy.stats import ranksums
res = ranksums(a1, a2)
print(res)
>>>RanksumsResult(statistic=-1.7320508075688772, pvalue=0.0832645166635504)
```


Wilcoxson signed-rank test

Very simple test: sort the data of the groups and add up the positions (rank) of each one when it is higher than the other.

Idea: the ranks of the sorts in which one of the two groups exceeds the other are added in turn. The more different the sums, the more different the groups will be. If similar groups, the high and low positions in the sorting will be equally divided.

The Wilcoxon W test statistic is the smaller of the two sums. The lower W the less likely it is that the difference is due to chance. (in fact, if low one of the two is always better).

A table of critical values indicates the probability of obtaining any particular value of W only by chance. (Note: Wilcoxon is atypical. Usually, the higher the statistic, the lower the prob. to have had it by chance. Here the inverse).

Parametric and nonparametric tests

	Nonparametric tests		Parametric Tests
	Nominal data	*Ordinal data*	*Ordinal data, range, ratio*
One group	Chi square goodness of fit	Wilcoxon signed rank	t test single group
Two unrelated groups	Chi square	Wilcoxon rank sum, Mann-Whitney test	t test two groups
Two related groups	McNemar's	Wilcoxon signed rank	Paired t test
n unrelated groups	Chi squared	Kruskal -Wallis, Analysis of variance	ANOVA
N related gropus		Friedman	ANOVA with repeated measurements

Feature selection

Feature Selection asks to select the features which are relevant to model. Only the attributes that have a significant effect on the model's output are selected.

In real cases, a dataset can have thousand of features and some of them may be ***redundant***, some may be ***correlated*** and some may be ***irrelevant*** for the model.

The use of all the features will take a lot of time to train the model, and model accuracy will be reduced.

There are many approaches to feature selection, including recursive feature elimination, Bayesian networks, decision trees, etc. Here only the most basic and manual method of filtering using statistical tests.

Examples on the **IBM's loan prediction dataset**. Used to make a predictive model to predict if an applicant is able to repay a lending company or not.

`https://github.com/mridulrb/Predict-loan-eligibility-using-IBM-Watson-Studio/blob/master/Dataset/Dataset.csv`

Many features in the dataset: Gender, Dependents, Education, Applicant Income, etc. We check whether one feature affects other ones using several tests: t-Test, correlation test, ANOVA test, and Chi-square test.

Feature selection: t test

A t-test is used to compare the mean of two given samples, even when the sample size is less than 30. It ***assumes a normal distribution*** of the sample.

In the example t-test where one variable - *Gender* - is categorical and separates the two samples and the other variable - *ApplicantIncome* - is continuous and checked for mean difference.

Null Hypothesis: There is no significant difference between the mean income of males and females.

Alternate Hypothesis: There is a significant difference between the mean income of males and females.

Feature selection: t test

```
from scipy.stats import sem,t
# function for calculating the t-test for two independent samples
def ttest(data1, data2, alpha):
    mean1, mean2 = mean(data1), mean(data2)    # means
    se1, se2 = sem(data1), sem(data2)          # standard errors
    sed = np.sqrt(se1 ** 2.0 + se2 ** 2.0)     # standard error on diff between samples
    t_stat = (mean1 - mean2) / sed             # t statistic
    degf = len(data1) + len(data2) - 2         # degrees of freedom
    cv = t.ppf(1.0 - alpha, degf) #critical value, ppf percent point function inv. of cdf
    p = (1.0 - t.cdf(abs(t_stat), degf)) * 2.0 # p-value
    return t_stat, degf, cv, p

df = pd.read_csv("../../serie/loanPrediction.csv")
# t test function call
data1 = df.loc[df['Gender'] == 'Male',   'ApplicantIncome']
data2 = df.loc[df['Gender'] == 'Female', 'ApplicantIncome']
alpha = 0.05
t, degf, cv, p = ttest(data1, data2, alpha)
```

We accept the null hypothesis since value p is greater than 0.05.
There is no significant difference between the income of males and females.

Feature selection: ANOVA

ANOVA, ANalysis Of VAriance, uses variance as its parameter to compare multiple independent groups.
ANOVA can be one-way ANOVA or two-way ANOVA.
One-way ANOVA is applied when there are three or more independent groups of a variable.

Example
We will use the *Dependents* categorial variable and the *ApplicantIncome* continuous variable.
Dependents has 4 groups (0,1,2,3+). The hypothesis are:
Null Hypothesis: there is no significant difference between income among different numbers of applicant's dependents.
Alternate Hypothesis: there is a significant difference between income among different numbers of applicant's dependents.

Feature selection: ANOVA

```
# missing values in the (applicant's) Dependents feature.
numna = df['Dependents'].isnull().sum()
df['Dependents'] = df['Dependents'].fillna('0')
df_anova = df[['ApplicantIncome', 'Dependents']] # operational dataframe
dval = pd.unique(df.Dependents.values)  # unique dependents numbers, will be the groups
# a dictionary with the four groups of appl/dep pairs
d_data = {val: df_anova['ApplicantIncome'][df_anova.Dependents == val] for val in dval}
# anova based on F statistics
F, p = stats.f_oneway(d_data['0'], d_data['1'], d_data['2'], d_data['3+'])
print('F ={},p={}'.format(F, p))
if p < 0.05: print("reject null hypothesis")
else:        print("accept null hypothesis")
```

F =5.955112389949444,p=0.0005260114222572804
Value p is less than 0.5, we reject the null hypothesis.
We conclude that there is a significant difference given by whether the applicant has dependents or not.

Feature selection: chi square

The Chi-Square (χ^2) test is applied on two **categorical** variables in order to determine whether there is a significant relationship between them.
Two types of chi-square tests: chi-square **goodness of fit** and chi-square **test for independence**.

The Formula for Chi Square Is

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

c = degrees of freedom

O = observed value(s)

E = expected value(s)

The *degree of freedom* in the chi-square test is calculated by (n-1)*(m-1) where n and m are numbers of rows and columns respectively.

Example

Test for independence. Categorical features *Gender* and *loanStatus*.

Null Hypothesis: There is no significant association between *gender* and *loanStatus*.

Alternate Hypothesis: There is a significant association among them.

Feature selection: chi square

```
# feature matrix via crosstab (frequency table of the factors)
dataset_table = pd.crosstab(dataset["Gender"], dataset["Loan_Status"])
# chi2_contingency() yields the p value for rejecting or accepting null hypothesis
chi_square, p, dof, expected = stats.chi2_contingency(dataset_table)
print("chi-square statistic:-", chi_square)
print('Significance level: ', alpha)
print('Degree of Freedom: ', dof)
print('p-value:', p)
if p <= alpha: print("Reject Null Hypothesis")
else:          print("Accept Null Hypthesis")
```

```
Gender/Loan_Status  N  Y
Female             37   75
Male               33  339
chi-square statistic:- 0.13962612116543877
Significance level:  0.05
Degree of Freedom:  1
p-value: 0.7086529816451106
Accept Null Hypthesis
```

Since the p-value is greater than 0.05, we accept the null hypothesis.
There is no significant association between the two features

Feature selection: Pearson

Correlation tests evaluate the extent to which continuous variables are related to one another.

Several methods for correlation tests: covariance, Pearson correlation coefficient, Spearman rank correlation coefficient, etc.

Pearson Correlation Coefficient r is a parametric test used to measure the linear correlation between 2 variables. Its values lie between -1 and 1.

- If $r = 0$, there is no relationship between variables X and Y.
- If $0 < r < 1$, there is a positive relation between X and Y, and their strength increases from 0 to 1: if the value of X increases, the value of Y also increases.
- If $-1 < r < 0$, there is a negative relation between X and Y, and their strength decreases from -1 to 0: if the value of X increases, the value of Y decreases.

In the example we use two continuous features – *loanAmount* and *applicantIncome*. There are missing values in the second.

Feature selection: Pearson

```
# fill missing values with the mean value.
df["LoanAmount"]=df["LoanAmount"].fillna(df["LoanAmount"].mean())
# calculate Pearson correlation coefficient
pcc = np.corrcoef(df.ApplicantIncome, df.LoanAmount)
print(f"r={pcc}")
plt.plot(df['LoanAmount'],df['ApplicantIncome'],marker='.',linewidth=0)
plt.show()
```

Result, r=[[1. 0.56562046]
[0.56562046 1.]]

The values on the diagonals indicate the correlation of features with themselves.

0.56 shows that there is some correlation between the two features.

Comparison: Diebold Mariano

Several plausible methods to forecast a time series, differences in forecast accuracy between the methods. Is this outcome due to pure chance, is the difference statistically significant or not?

If there are just two plausible models there are model-free tests (do not consider the models that generated the forecasts) that compare the forecasts, for example Diebold-Mariano (DM, 1995), but more advanced tests exist.

They all use a null hypothesis H0 stating that the two forecasts have the same accuracy.

We skip the math, just an example with DM two-tails test, knowing that the test statistics DM is normal distributed. H0 is rejected if the DM statistic falls outside the range $\left[-z_{\alpha/2}, z_{\alpha/2}\right]$, that is if $|\mathrm{DM}| > z_{\alpha/2}$, where $z_{\alpha/2}$ is z-value from the standard normal distribution corresponding to half of the desired α level of the test.

Diebold Mariano

Example, suppose that the significance level is α = 0.05.

Since DM is a two-tailed test, 0.05 must be split into 0.025 for the upper tail and 0.025 for the lower tail.

The z-value corresponding to -0.025 is -1.96, this is the lower critical z-value. The upper critical value is computed for 1-0.025 = 0.975, which (unsurprisingly) has a z-value of 1.96.

The null hypothesis H0 will be rejected if the DM statistic falls outside the range of [-1.96, 1.96].

Diebold Mariano

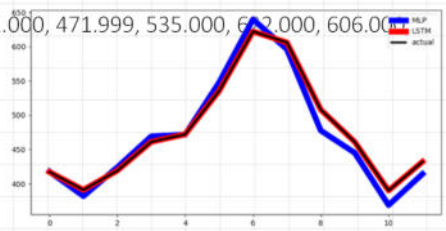
Example, using dm_test from https://github.com/johntwk/Diebold-Mariano-Test.

Compare MLP (200 epochs) with LSTM (100 epochs) on the Box Jenkins dataset, forecast last 12 periods with a 1 period lookahead per record.

Actual (out-of-sample) values:
[417,391,419,461,472,535,622,606,508,461,390,432]

MLP = [414.231, 417.662, 381.815, 423.952, 469.055, 471.502, 547.532, 639.609, 596.040, 477.471, 445.010, 368.674]

LSTM= [416.999, 390.999, 418.999, 461.000, 471.999, 535.000, 622.000, 606.000, 507.999, 461.000, 390.000, 431.999]



Diebold Mariano

```
from dm_test import dm_test
if __name__ == '__main__':
    MLP = [417.662,381.815,423.952,469.055, 471.502, 547.532, 639.609,
596.040, 477.471, 445.010, 368.674,414.231]
    LSTM= [416.999,390.999,418.999,461.000, 471.999, 535.000, 622.000,
606.000, 507.999, 461.000, 390.000, 431.999]
    actual = [417,391,419,461,472,535,622,606,508,461,390,432]
    rt = dm_test(actual, MLP, LSTM, h=1, crit="MSE")
    print(rt)
```

Mean squared error loss

One period lookahead when forecasting

Result:

```
dm_return(DM=2.933300449662608, p_value=0.013609843048263027)
```

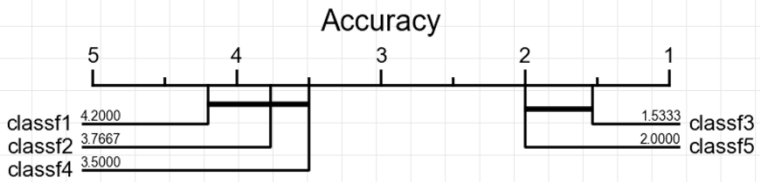
< 0.025, reject H0

> 1.96, reject H0

Critical difference diagrams

It is possible to include in a single diagram the results of several pairwise comparisons (`import scikit_posthocs as sp` or source).

The diagram below, based on Wilcoxon-Holm signed rank tests from the sources in `critical_difference_diagram.py`, shows the ranks in the ordering of 5 classes.

Classes 3 and 5 performed best, with no significant difference between them.

Classes 1, 2 and 4 perform worse and are equivalent among them.